Day 4 – Your First Lightning Web Component (LWC)

Part 1 – Think Before Coding

1. Why do users need a graphical interface?

A Graphical User Interface (GUI) allows users to interact with Salesforce easily without requiring programming or database knowledge. Instead of writing commands, users can perform tasks by clicking buttons, filling out forms, viewing records, and searching data, making the application simple and user-friendly.

2. Why can't users directly execute SOQL queries?

Regular users should not execute SOQL queries directly because they may not understand the query syntax, incorrect queries can lead to errors or poor performance, and direct database access does not follow business rules and security policies. Instead, the application retrieves the required data and displays it through the user interface.

3. Why is JavaScript required in LWC?

JavaScript controls the functionality and behavior of Lightning Web Components. It handles user interactions, stores and processes data, validates input, communicates with Apex methods, and updates the user interface dynamically. While HTML defines the structure of the component, JavaScript manages its logic.

4. What responsibilities belong to the UI?

The User Interface (UI) is responsible for presenting information and interacting with users. It displays records, forms, buttons, tables, and messages, accepts user input, performs basic client-side validation, sends requests to Apex when required, and displays success or error notifications.

5. Which responsibilities should remain in Apex?

Apex is responsible for server-side processing and business logic. It executes SOQL queries, performs database operations such as insert, update, and delete, implements business rules, processes large datasets, manages transactions, enforces security and data access, and provides data to Lightning Web Components.

Part 2 – Understanding LWC

A Lightning Web Component (LWC) consists of three main files:

1. studentHome.html – HTML

This file defines the user interface of the component. It contains the page layout, headings, buttons, text, images, input fields, and displays data that users interact with.

2. studentHome.js – JavaScript

This file contains the component's logic and behavior. It manages variables, handles events, processes user input, communicates with Apex methods, and controls actions such as button clicks.

3. studentHome.js-meta.xml – Meta XML

This configuration file defines where the component can be used within Salesforce. It exposes the component to Lightning App Builder and specifies targets such as App Pages, Home Pages, and Record Pages.

Easy Way to Remember

File	**Responsibility**
studentHome.html	Displays what the user sees
studentHome.js	Controls what the component does
studentHome.js-meta.xml	Defines where the component can be used

Part 3 – Hands-on Activity 1

Objective

Create the first Lightning Web Component (LWC) for the Placement Management System and deploy it on a Lightning Page.

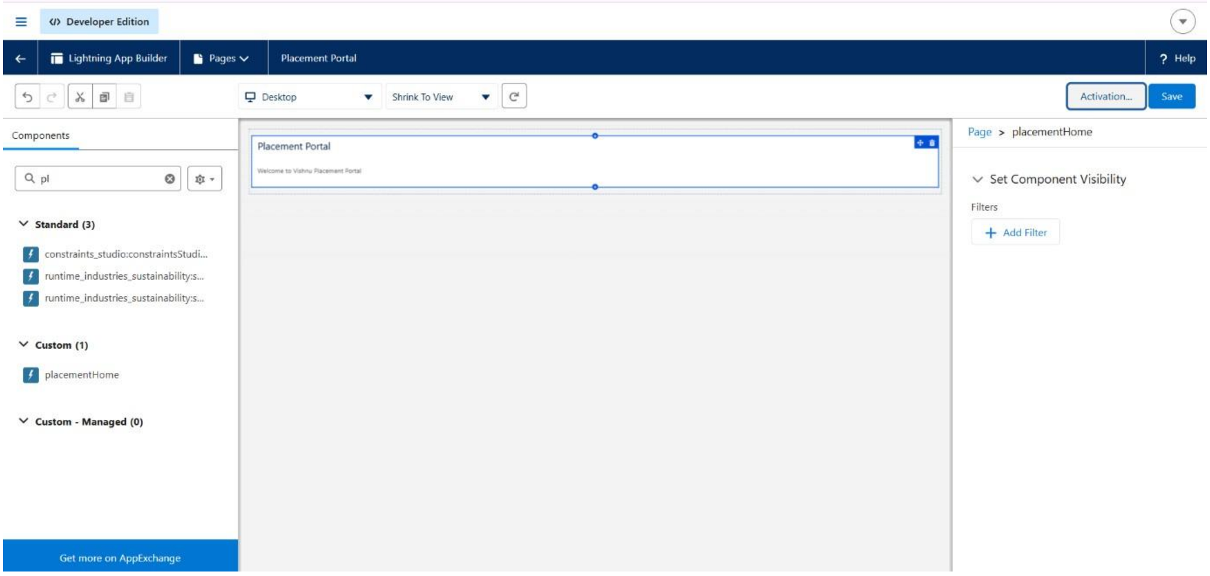
Component Name

placementHome

Expected Output

Placement Portal

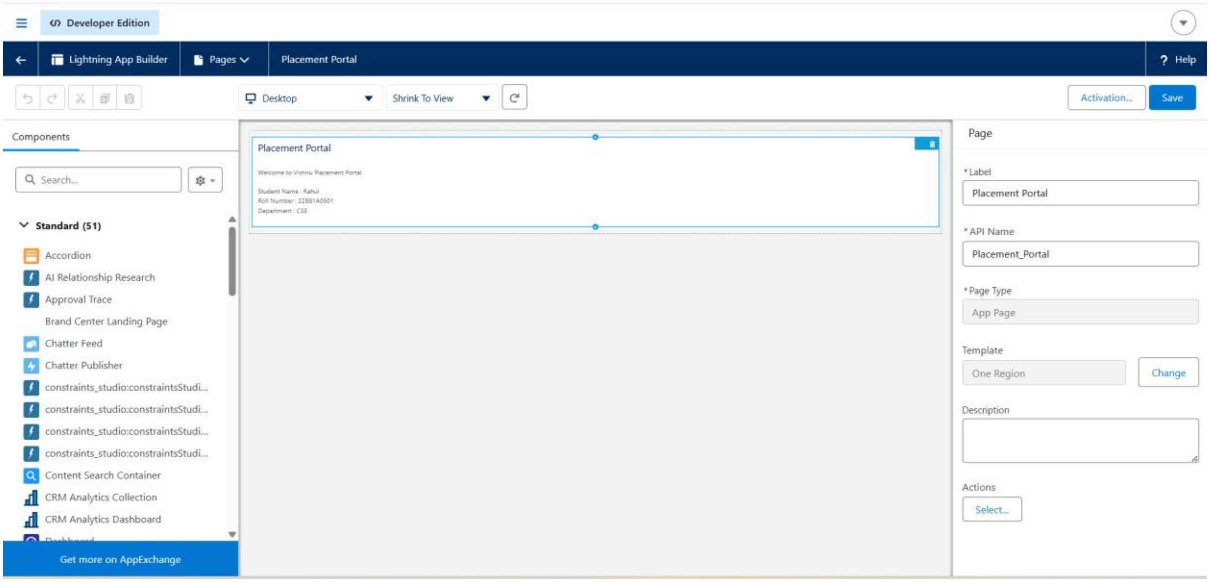
Welcome to Vishnu Placement Portal.



Part 4 – Hands-on Activity 2

Objective

To create Student Name, Roll Number, and Department variables in an LWC, display their values on the screen, and observe how the UI changes when the variable values are updated.



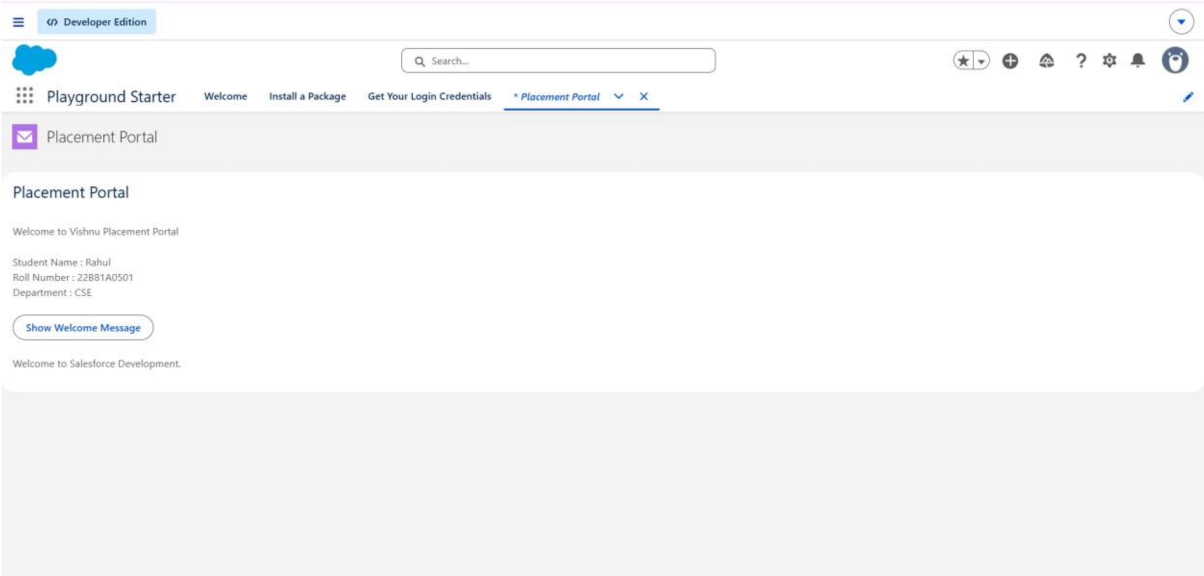
Part 5 – Hands-on Activity 3

Objective

To create a button in an LWC, handle a click event using JavaScript, and display a welcome message dynamically.

Initially, only the Show Welcome Message button appears.

After clicking the button, it displays: Welcome to Salesforce Development.



Part 6 – Hands-on Activity 4

Objective

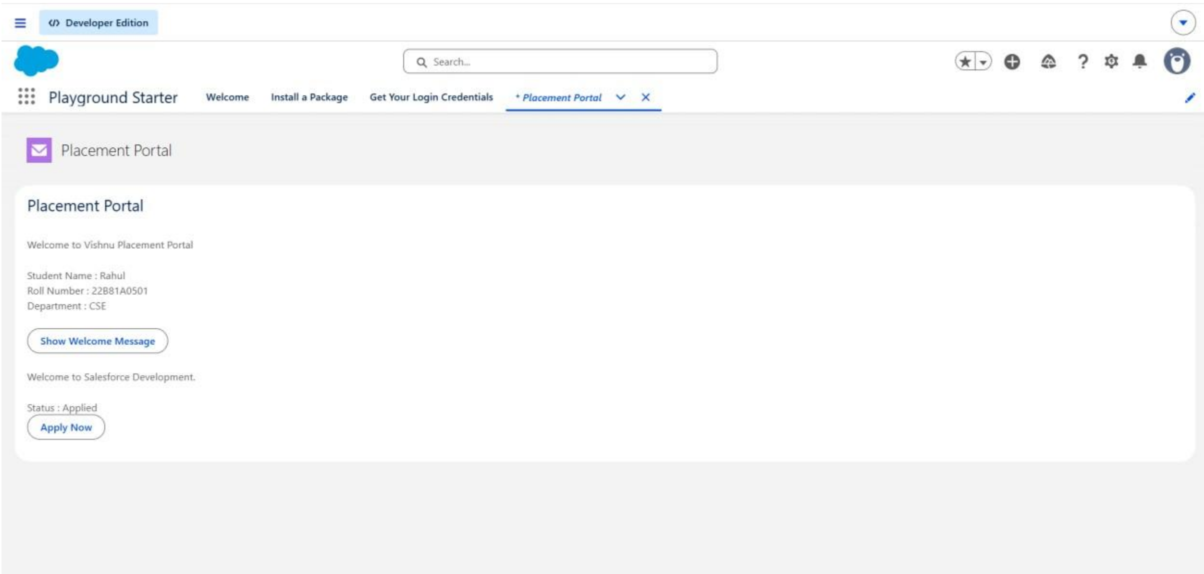
To understand how JavaScript can dynamically change data in an LWC without using Apex or a database.

Before clicking the button:

Status : **Not Applied**

After clicking Apply

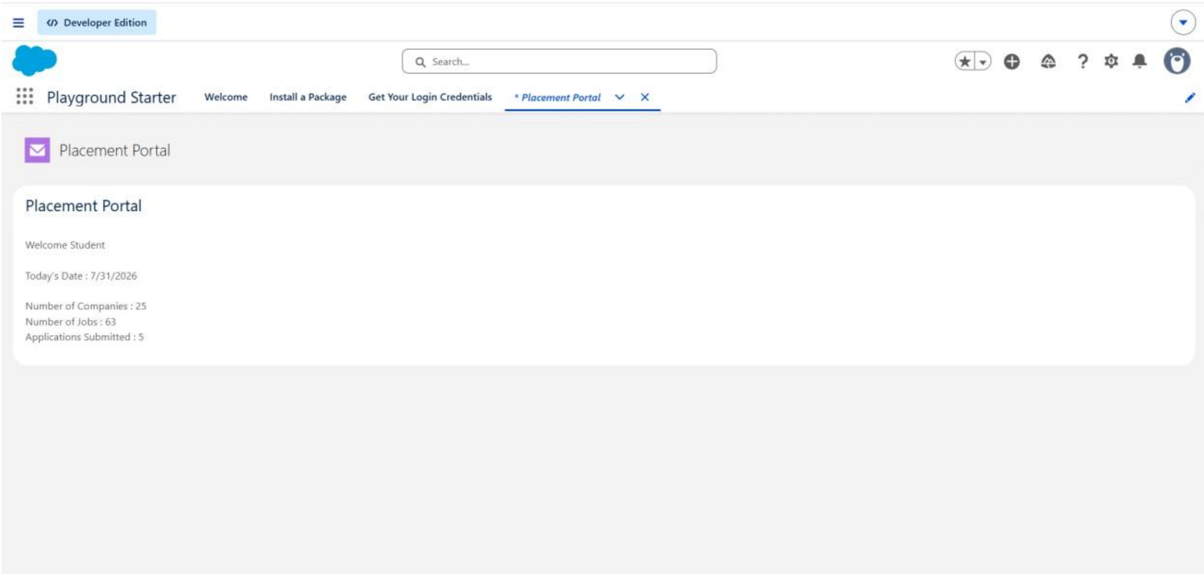
Status : **Applied**



Part 7 – Mini Project Enhancement

Objective

To create the first screen of the Placement Management System using LWC and display placement information using hard-coded JavaScript values.



Part 8 – Understanding Data Binding

Objective

To understand data binding in LWC by displaying a JavaScript variable in HTML and observing how the UI changes when the variable value is changed.